

Audio Context Layer for Audio Question Answering

Technical Report

1. Problem Formulation

The goal is to answer natural language questions about an audio recording while keeping the reasoning grounded in observable audio evidence. The system uses three stages: audio perception, an intermediate context layer, and question answering.

The perception model first produces timestamped event information. The context layer stores these detections in an inspectable JSON representation. A language model then reasons over the structured representation and returns an answer, confidence, and supporting event IDs.

2. Research Study

The project investigates whether an explicit audio context layer can provide a useful and interpretable interface between audio perception and language reasoning. AST, PANNs CNN14, YAMNet, and a small supervised CNN were evaluated.

YAMNet was selected for the downstream proof of concept based on the strongest measured micro and macro F1 and higher recall among the evaluated perception models.

3. Dataset

ESC-50 was used as the source audio collection. A smaller curated synthetic scene dataset was constructed from the source clips.

The final dataset contains 150 scenes: 100 training, 25 validation, and 25 test scenes. Scene duration ranges from 20 to 30 seconds, with 5 to 7 events per scene. All scenes contain overlapping events.

The QA dataset contains 1,200 pairs, with 8 questions per scene. Question types are apparent, causal reasoning, comparison, counting, duration, temporal, and yes/no.

ESC-50 folds 1 to 3 were used for training, fold 4 for validation, and fold 5 for testing to reduce source recording leakage.

4. Method and Design Decisions

The context layer stores event ID, label, start time, end time, duration, confidence, and source frame information.

Final YAMNet context settings: confidence threshold 0.30, minimum event duration 0.40 seconds, and maximum merge gap 1.50 seconds. The merge gap was increased from 0.75 to 1.50 seconds after spot checks showed that confidence dips could fragment sustained events.

The QA model receives only the structured context and the question. It is instructed not to assume events absent from the context and to return 'Cannot determine from the available audio context.' when evidence is insufficient.

5. Experimental Setup

Pretrained perception models were evaluated on 25 held out test scenes. The final benchmark contained 248 audio windows. Precision, recall, and F1 were measured at micro and macro levels.

The custom CNN used 5 second windows with a 2.5 second hop, 64 bin log Mel features, four convolutional blocks, weighted BCEWithLogitsLoss, Adam, 15 epochs, and seed 42.

The downstream QA evaluation contains 200 test questions. Objective scoring covers apparent, comparison, counting, duration, temporal, and yes/no questions. Causal reasoning is reported separately using an environment level heuristic.

6. Results

Perception results: AST micro F1 0.4198 and macro F1 0.3773; PANNs CNN14 micro F1 0.4822 and macro F1 0.4311; YAMNet micro F1 0.5681 and macro F1 0.5214; Custom CNN micro F1 0.3811 and macro F1 0.3708.

QA results: apparent F1 0.6621; causal reasoning environment accuracy 0.9600; comparison 0.0800; counting 0.3200; duration 0.2000; temporal 0.5000; yes/no 0.7600; overall objective accuracy 0.3543.

7. Error Analysis

There were 114 incorrect scored predictions out of 200. A rule based offline attribution classified 98 cases (86.0%) as perception or context errors, 3 cases (2.6%) as QA reasoning errors, and 13 cases (11.4%) as mixed or ambiguous.

Typical perception/context failures included missing events, merged or incomplete event intervals, incorrect counts, and duration mismatches. This attribution is heuristic rather than causal proof.

8. Loss Curve

The custom CNN training loss curve is stored in results/training_loss.png. The trained checkpoint is stored in models/audio_cnn_best.pt.

9. Observations and Limitations

The dominant bottleneck is audio perception fidelity, especially missed events and inaccurate temporal boundaries. Downstream reasoning cannot recover information absent from the context layer.

The dataset is small and synthetic. Scenes are constructed from short source clips rather than naturally recorded continuous environments. Causal questions cannot establish real world causality from audio alone.

The perception benchmark and frame level context construction use different temporal aggregation strategies. A larger follow up experiment should align these operating points.

10. Conclusion

The project demonstrates an audio question answering pipeline built around an explicit intermediate context representation. The context layer makes the system easier to inspect because answers can be connected to timestamped detected events.

The experiments indicate that improving event detection and temporal reconstruction is an important direction for improving downstream QA.

11. Reproducibility

The repository contains scene manifests, QA data, generated context examples, evaluation results, training code, perception evaluation scripts, QA scoring scripts, and the trained custom CNN checkpoint. Original ESC-50 audio and generated scene WAV files are excluded to keep the repository small.

The QA implementation reads GROQ_API_KEY from the environment. API credentials are not stored in the repository.